The Bilinear GCIQL Critic Is Not a Learned Temporal Metric

An established negative on OGBench cube, and a mathematical account of what the critic is doing instead

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Scope. This note isolates one result of the decoding-bilinear project: the refutation of the *metric hypothesis* for bilinear GCIQL (BIQL) on OGBench `cube-single-play-v0`, with every number from our own probe matrix (3 seeds × 500k steps per condition; probe battery of `docs/specs/2026-07-31-bilinear-gciql-metric-probe.md`). It ends (§3) with the mathematical argument for the surviving alternative. Companion notes: `phenomenon.tex` (pre-registered phenomena and kill-tests), `decoding-bilinear.tex` (kernel formalism).

1 The hypothesis under test

The critic is $\hat{Q}(s,a,g) = \langle \varphi(s,a), \psi(g) \rangle / \sqrt{d}$ with $\hat{V}(s,g) = \langle \varphi(s), \psi(g) \rangle / \sqrt{d}$, trained by goal-conditioned expectile TD ($\tau = 0.9$, $\gamma = 0.99$) on relabeled play data. Let $d^*(s,g)$ denote the minimal dataset-supported hitting time and, for any critic construction, let $\hat{d}(s,g)$ be its induced distance (for BIQL: $\hat{d} = \log_\gamma(1 + (1-\gamma)\hat{V})$, or the self-subtracted logit $f(g,g) - f(s,g)$; both are monotone reparametrizations of $\hat{V}$ and rank-equivalent for every probe below).

Hypothesis 1 (Metric hypothesis, H-metric)**.** *BIQL's policy advantage over the monolithic GCIQL (*92.7 ± 1.4 *vs.* 73.2 ± 1.3 *success at 500k; pure-BC floor* 8–10*) is mediated by the* fidelity of its learned geometry*: $\hat{d}_{BIQL}$ approximates d^* (equivalently, embeds a temporal metric) better than $\hat{d}_{GCIQL}$, and this fidelity causes the better guidance.*

If H-metric holds, BIQL must (a) rank temporal distances at least as well as GCIQL on held-out pairs, (b) agree geometrically with the architecturally metric-constrained references (QRL; and CRL's contrastive embedding), and (c) improve its geometry over the training window in which success improves.

2 The evidence: all three implications fail

Finding 1 (Temporal ranking: GCIQL is the better metric, and loses)**.** *Probe P1, Spearman* $\rho(\hat{d}, \Delta t)$ *over* 10^5 *same-trajectory validation pairs (second number: partial correlation controlling raw-observation* L_2*):*

$$\textit{GCIQL}: \ 0.63 \,/\, 0.54 \qquad > \qquad \textit{BIQL}: \ 0.54 \,/\, 0.41.$$

The critic with the worse *temporal geometry wins by 20 success points. Fidelity of* $\hat{d}$ *to* d^* *on the data distribution is therefore not sufficient (GCIQL) and not necessary (BIQL) for guidance value.*

Finding 2 (Representational agreement: BIQL is the geometric outlier)**.** *RSA (Spearman between upper triangles of the* 2000×2000 *native distance matrices):* $\mathrm{RSA}(\textit{GCIQL}, \textit{QRL}) = 0.88$*,* $\mathrm{RSA}(\textit{GCIQL}, \textit{CRL}) = 0.54$ *— a three-method* consensus geometry *among the metric-flavoured methods. BIQL agrees with none of them (*≤ 0.21 *against every learned method) and sits closest to* raw observation distance *(*0.45–0.52*). If BIQL had learned the temporal metric, it would join the consensus; instead the consensus members underperform it by 20–60 points.*

Finding 3 (The cosine structure is architectural, not learned)**.** *The trained head is numerically a scaled cosine (*$\mathrm{corr}(\langle \varphi, \psi \rangle, \cos \angle(\varphi, \psi)) = 0.996$*; norm CVs* ≤ 0.09*) — but the* untrained *network already has cos-correlation* 0.994*: LayerNorm concentrates tower norms at initialization. "It learned a cosine similarity" is a property of the architecture class, not of training, and cannot explain a training-acquired advantage.*

Finding 4 (Temporal dissociation: geometry saturates, success does not)**.** *From 100k to 500k steps,* BIQL*'s P1-partial moves* $0.37 \to 0.41$ *while success moves* $65 \to 93$*. The quantity H-metric says is causal is flat over the window in which the effect it must explain triples.*

Finding 5 (Where fidelity actually differs: eval-task rollouts, with the sign reversed)**.** *Probe P2 (2026-08-03): per-episode Spearman* $\rho(\hat{V}(s_t, g),\, t)$ *along temperature-0 eval rollouts, full* 2×2 *of scoring critic* $\times$ *rollout-generating policy (3 seeds* $\times$ *50 episodes; median over episodes, mean*$\pm$*sd over seeds):*

scored on $\to$	BIQL*-policy rollouts (*0.95 *succ.)*	GCIQL*-policy rollouts (*0.73 *succ.)*
BIQL *critic*	0.983 ± 0.003	0.871 ± 0.060
GCIQL *critic*	$\mathbf{0.355 \pm 0.239}$	0.883 ± 0.015

On val-set pairs GCIQL *ranks better (Finding 1); on the eval-task states of the* successful *policy its value signal is noise (per-task medians as low as* -0.80*), while* BIQL*'s is a near-perfect monotone ramp. Each critic is coherent along its own policy's trajectories (diagonal); only* BIQL transfers *to the other policy's state distribution. So the fidelity that discriminates the two critics is not metric fidelity on the data distribution — it is* off-distribution coherence *on exactly the* (s, g) *pairs eval interrogates.*

Remark 1 (Verdict)**.** *H-metric is refuted on every implication: (a) fails by Finding 1, (b) by Findings 2–3, (c) by Finding 4; and Finding 5 locates the real discriminator elsewhere.* BIQL *also violates metric axioms freely where it wins (asymmetry index and triangle-violation rates comparable to* GCIQL*'s in the trust region, magnitudes bounded) — metric* axioms *are neither enforced nor selected for. Whatever* BIQL *learns, calling it a metric explains none of the observed orderings.*

3 What the critic is plausibly doing instead

The surviving account is that the bilinear head is a **bounded low-rank completion of the in-sample-optimal reachability kernel**, and that its advantage is a property of the *function class at the actor interface*, not of any geometry it encodes. We give the argument in three steps, each attached to a measured quantity.

Setup. Let $K^*(s, g) = \gamma^{d^*(s,g)} \in [0, 1]$ and let $V^* = (K^* - 1)/(1 - \gamma)$ be the optimal goal-conditioned value under the -1-per-step convention, so $V^* \in [-1/(1-\gamma),\, 0] = [-100, 0]$ at $\gamma = 0.99$. Expectile TD at $\tau \to 1$ regresses toward the in-sample optimal V^* on the relabeled pair measure μ. The bilinear class with LayerNorm towers is

$$\mathcal{F}_d \;=\; \Big\{(s, g) \mapsto \tfrac{1}{\sqrt{d}} \langle \varphi(s), \psi(g) \rangle \;:\; \|\varphi\| \approx \kappa_\varphi,\; \|\psi\| \approx \kappa_\psi \Big\}, \qquad \kappa := \kappa_\varphi \kappa_\psi / \sqrt{d}.$$

Measured on BIQL-d50: $\kappa \approx 75$, realized $\hat{V} \in [-63, +6]$.

Step 1 (boundedness caps actor exploitation). The one-step actor maximizes $\hat{Q}$ against a BC anchor. For any policy π reached by ascent, the standard decomposition gives

$$J(\pi) \;\geq\; J(\pi_{\mathrm{BC}}) \;+\; \underbrace{\mathbb{E}_{s,g}\big[\hat{Q}(s, \pi) - \hat{Q}(s, \pi_{\mathrm{BC}})\big]}_{\text{what ascent collects}} \;-\; 2 \sup_{s,a,g} \big|\hat{Q} - Q^*\big|.$$

For an unbounded monolithic MLP the sup is uncontrolled precisely off the data distribution, where ascent goes hunting; within $\mathcal{F}_d$ it is at most $\kappa + \sup|Q^*|$, i.e. the class cannot manufacture value spikes: since $|\hat{Q}| \leq \kappa \approx 75 < 100 = \sup|V^*|$, the class cannot even *represent* the most extreme values, let alone hallucinate beyond them. Errors are forced to be *many, small, and spread* rather than rare and enormous — exactly the measured triangle-violation profile (BIQL magnitude 4.8 steps vs. GCIQL 9.5 at comparable rates).

Step 2 (rank d forces completion to unseen pairs). Write the value table on supports $\mathcal{S}\times\mathcal{G}$ as a matrix $M_{sg} = \hat{V}(s,g)$; $\mathcal{F}_d$ constrains $\mathrm{rank}(M) \le d$. The training measure μ covers a sub-block pattern $\Omega \subset \mathcal{S}\times\mathcal{G}$ (each state with *some* goals, each goal with *some* states — play relabeling guarantees marginal coverage, not joint coverage). A rank-d matrix has $d(|\mathcal{S}| + |\mathcal{G}| - d)$ degrees of freedom, and under standard incoherence conditions is *uniquely determined* by $O\big(d\,(|\mathcal{S}| + |\mathcal{G}|)\,\mathrm{polylog}\big)$ generically positioned entries. Hence, once μ pins the factors on Ω, the values on the *unmeasured* block $(\mathcal{S}\times\mathcal{G})\setminus\Omega$ — which contains the eval pairs (s, g^{task}) — are not free parameters: the class extrapolates them by completion, $\hat{V} = \Phi\Psi^\top/\sqrt{d}$ with both factors identified on the covered marginals. A monolithic MLP obeys no such constraint; its off-Ω values are limited only by smoothness. This is Finding 5 in mathematical form: the 2×2 transfer asymmetry (coherent at 0.87 off its training policy's distribution vs. GCIQL's 0.36) is the completion property, observed.

Step 3 (what is learned is a spectral object, not a metric). The expectile regression solves, in the idealized linear-algebraic limit,

$$\min_{\mathrm{rank}\,M\le d} \big\|W_\tau \odot (M - V^*)\big\|_\mu^2,$$

a τ-weighted low-rank approximation of V^*: the learned object is (a reweighting of) the **top-d spectral subspace of the optimal value kernel**, not a distance function. This predicts, and the data confirm:

(i) performance depends on capturing the top spectral subspace, so success saturates once d exceeds the kernel's effective spectral dimension: measured saturation at $d^\dagger \in [10, 20]$ (0.06/0.62/0.92/0.90/0.93 at $d = 5/10/20/30/50$), with learned effective rank tracking the container sublinearly (4.1/8.0/14.7/19.6/28.0) — a capacity budget, not a learned constant;

(ii) metric axioms (symmetry, triangle inequality) are epiphenomenal — a truncated spectral approximation of γ^{d^*} satisfies neither exactly, and BIQL indeed violates both while winning;

(iii) geometry probes on μ (P1, RSA) can prefer GCIQL while the actor prefers BIQL, because they measure pointwise ranking on Ω, which is blind to both boundedness (Step 1) and off-Ω completion (Step 2).

Proposition 1 (Value-completion account, falsifiable summary)**.** *BIQL's guidance advantage = bounded ($\kappa < \sup|V^*|$), rank-$d^\dagger$ completion of the expectile-optimal reachability kernel from marginal coverage. It predicts: (a) bounding a monolithic critic at κ recovers part of the $73 \to 93$ gap (`gciql-qb75`, running); (b) destroying norm concentration degrades BIQL at fixed rank (`biql-d50-nopsiln`, running); (c) the completion behaviour of Finding 5 should survive any monotone reparametrization of $\hat{V}$ and vanish for $d < d^\dagger$. Any of these failing bounds how much of the account survives; none of them resurrects H-metric, which is refuted by Findings 1–5 independently.*